

Data Preparation

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3.1 Selecting Data

The selected modeling features are region, price, sqfeet, beds, baths, cats_allowed, dogs_allowed, smoking_allowed, wheelchair_access, electric_vehicle_charge, comes_furnished, laundry_options, parking_options, lat, long, and state. Four columns were dropped from the working dataset: id, url, image_url, and description. These excluded fields carry no predictive signal for price modeling. The type column was retained during selection but dropped after the apartment-only filter in Task 3.2 (Cleaning), since it becomes a single-value column with no modeling utility.

3.2 Cleaning Data

Records with $\text{price} < 1$ and $\text{sqfeet} < 70$ were removed as implausible values. Listing-level duplicate checks confirmed zero duplicate rows in the dataset.

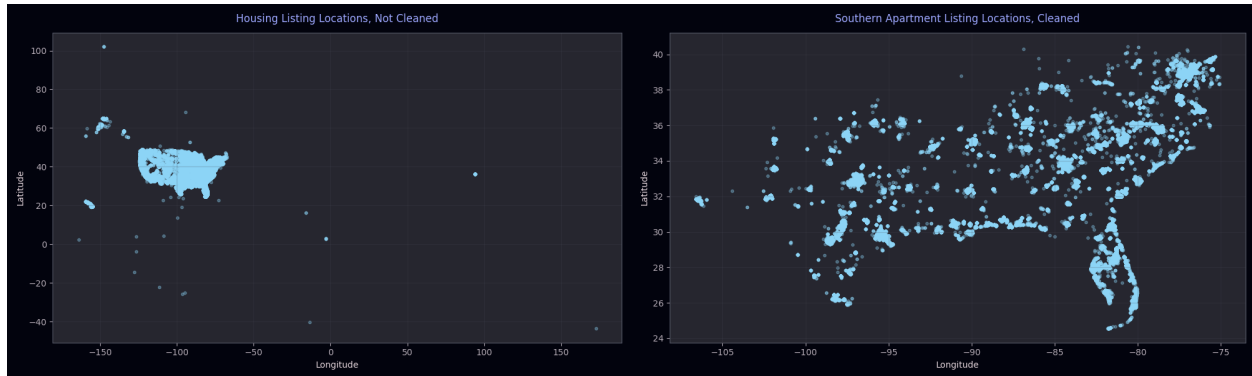
Geographic filtering was applied next. A single bounding box covering the 17 Southern states was applied directly to `df_selected`: latitude $\text{SOUTH_LAT_MIN} = 24.55^\circ$ (Key West, FL) to $\text{SOUTH_LAT_MAX} = 40.64^\circ$ (WV Northern Panhandle), and longitude $\text{SOUTH_LONG_MIN} = -106.65^\circ$ (El Paso, TX) to $\text{SOUTH_LONG_MAX} = -75.05^\circ$ (MD/DE Atlantic coast).

The dataset was then filtered to the 17 Southern states per the Census Bureau definition (AL, AR, DC, DE, FL, GA, KY, LA, MD, MS, NC, OK, SC, TN, TX, VA, WV). Because the geographic bounding box was applied first, this second pass also catches any records with a valid Southern state code but coordinates that fell inside the box through mislabeling rather than actual location. The filtered data spans 148 unique Craigslist regions.

Finally, the dataset was restricted to apartments only, per the project scope definition. Since type becomes constant after this filter, the column was dropped. Figures 1a and 1b show the full dataset before cleaning and the Southern-apartment subset after all filters but before IQR-based outlier removal.

Price outliers were treated using the IQR method, applied per region. A per-region approach respects the distinct price distributions of different housing markets: a luxury outlier in rural

Figure 1
Housing Listing Locations



(a) *Not Cleaned*

(b) *Cleaned: Southern States (Apartments Only)*

Mississippi and a typical listing in a major metro area may share the same dollar figure, and a global rule would conflate them. The IQR filter is applied to the price column only, leaving other numeric features untouched to avoid deleting valid but uncommon listings (large homes, high bedroom counts, etc.).

No cleaning was applied to categorical features (`cats_allowed`, `dogs_allowed`, `smoking_allowed`, `wheelchair_access`, `electric_vehicle_charge`, `comes_furnished`, `laundry_options`, `parking_options`), as these are labels rather than numeric measurements and are not subject to outlier or range checks.

3.3 Constructing Data

A new feature, `location_cluster`, was engineered from latitude and longitude using K-Means clustering with $k = 148$ clusters, matching the number of unique Craigslist regions in the filtered Southern-states dataset. The intent was to group properties into geographic clusters that approximate proximity to local rental sub-markets, capturing pricing patterns that raw coordinates or state-level groupings alone cannot express.

Coordinates were imputed with the median prior to clustering. Cluster labels were converted to strings and treated as a categorical feature in downstream encoding. The data was split 80/20 into train and test sets (`random_state=42`) before clustering to prevent data leakage: the K-Means model was fit exclusively on training coordinates.

No separate data-generation step was performed beyond this feature construction.

3.4 Integrating Data

No data integration was required. The project uses a single, self-contained CSV dataset from Kaggle with no external tables, APIs, or supplementary sources to merge.

3.5 Formatting Data

Numeric features were imputed with the median. No scaling or normalization was applied, as the project uses tree-based models (Random Forest, HistGradientBoostingRegressor, XGBoost) that are invariant to monotonic transformations of numeric features.

Categorical features received two distinct treatments based on their missing-data patterns. For `laundry_options` and `parking_options`: missing values were replaced with the literal string "Missing", preserving the absence of information as a distinct category that may itself carry predictive signal (e.g., listings that omit parking details may differ systematically from those that declare “no parking”). For all other categorical features (`state`, `region`, `location_cluster`), missing values were filled with the mode.

All categorical features were encoded using `TargetEncoder` from the `category_encoders` library with `smoothing` set to 10. Target encoding was chosen over one-hot encoding because several categorical features have high cardinality (`state`: 17 levels, `region`: 148 Craigslist metro areas, `location_cluster`: 148 levels). One-hot encoding would have created hundreds of sparse binary columns, inflating training time and complicating feature importance interpretation, while ordinal encoding would have imposed a false ordering on nominal categories. Target encoding compresses each categorical level to a single numeric value (the smoothed mean of the target), preserving predictive signal without dimensionality explosion. This reduced per-model training time from an estimated 60–90 minutes to under one minute.

After preprocessing, the final training matrix contained 11 numeric features and 5 target-encoded categorical features; the test matrix was identically structured. No "Missing" or "value" string artifacts remained in either matrix after encoding, confirming that the preprocessing pipeline executed without leakage or data corruption.